

E.3 / PAPER 1 / QUESTION 21

25 October 2023 / TZ1 (file label) / HL Paper 1 / original Q19 / 8823-6507

21. A sample of the artificial element Seaborgium-265 (Sg-265) is created with 4.0×10^{22} atoms. Sg-265 has a half-life of 15 s. How many atoms of Sg-265 remain after one minute?

A. 2.5×10^{21}

B. 5.0×10^{21}

C. 1.0×10^{22}

D. 2.0×10^{22}